

Gregory Bian

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September 17, 2026

Hiring Manager
Tesla
Langley, British Columbia

Dear Tesla Hiring Manager,

I am writing to express my interest in the Robotics Test Automation Engineer, Optimus position at Tesla. I am a second-year computer engineering student at UBC with a passion for embedded systems and robotics. Tesla's work building the Optimus Actuator Testing team's validation pipeline for next-generation actuator gearboxes is particularly compelling to me, as it combines PLC-based test automation, electrical system bring-up, and data-driven analysis in a fast-paced hardware development environment. I am excited about the opportunity to write Structured Text and Python test routines that turn gearbox validation into a repeatable, reliable process. Through my coursework and projects, I have developed experience programming microcontrollers in C/C++, designing and reviewing circuits and schematics, and working with communication protocols such as I2C, SPI, and CAN. I am motivated to apply these skills in a fast-paced environment while learning from a multidisciplinary team pushing the boundaries of robotic hardware.

In my current role as an embedded firmware developer with UBC Solar Racing's design team, I have been working on developing firmware in C for an STM32 microcontroller to display driver and diagnostic data on an LCD screen. For this project, I interpreted STM32 pin-out diagrams to read and write data to the LCD, scheduled tasks concurrently with LCD updates using FreeRTOS, and read data over the CAN bus to be displayed on the LCD screen. After consulting with the Power and Signals and BMS subteams to ensure that my firmware met hardware constraints, I was able to improve visibility of relevant data shown to drivers by 30% and confirm reliable data transmission for diagnostic data to ensure driver safety. This work accelerated my knowledge of embedded systems and working within a team, making me well equipped to contribute to your team.

My leadership, teamwork, and communication skills are demonstrated in my previous role as an STEM and Computing/Robotics Instructor at UBC Geering Up Outreach. In this role, I taught week-long summer camps to up to 30 students each week on topics such as circuit design and programming with Arduino, computer architecture, and introduction to Python. For each lesson, I communicated effectively with my co-instructor to create interactive and engaging lesson plans, coordinated volunteers to gather necessary materials, and facilitated a fun and enjoyable learning experience for the students. As a result, I was able to keep camper incidents to a minimum resulting in positive feedback and satisfaction scores on post-camp surveys. I would be excited to apply these skills to this role while furthering my technical experience.

Beyond work, I enjoy working on robotics and automation side projects displayed on my [Portfolio](#). I would love the opportunity to bring my passion for robotics and collaborative spirit to Tesla. I am available for a 8-month term starting May 2026 and welcome the opportunity to discuss how my skills and experiences align with your needs. Please feel free to contact me through email at gregorybian100@gmail.com or by phone at 604-652-7598.

Thank you for your time and consideration. I look forward to the possibility of contributing to your team.

Sincerely,

Gregory Bian